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Geometric Sector Dynamics on a Cone: Effective Couplings, Multiplicity, and Open Microscopic Closures

Abstract

We present a consolidated account of a cone-sector dynamical framework intended to connect a static conical geometric background with effective winding-based interactions and asymptotic scaling behavior. The purpose of this paper is not to claim complete microscopic closure of the full program, but rather to preserve and organize the progress achieved so far across several layers of description. This paper preserves progress by distinguishing foundational geometric structure, reduced-model derivations, calibrated quantities, and open microscopic closures. We highlight two main preserved results: an exchange-odd mismatch coupling arising in a parent-kernel reduction, and an effective multiplicity $n = 8$ derived from a geometric winding estimate. We emphasize that these are model-level derivations, not yet full first-principles closures. Finally, we state the role of the Renormalization Group / Relativistic Dark Geometry (RG/RDG) extension as a proposed deeper derivation program.

1. Introduction

This paper consolidates the current state of a cone-sector dynamical program whose long-term goal is to relate conical geometric structure, winding-sector dynamics, and effective large-scale phenomenology within a single framework. The evolution of the Tensile Asymptotic Field Approximation (TAFA) framework has moved systematically from empirical phenomenology toward a comprehensive geometric model.

However, several pieces of the program have been developed at different levels of generality: some belong to the foundational geometry, some arise only after reduction to effective sector variables, some have been calibrated phenomenologically, and others remain open targets for a deeper derivational program. In this situation, it is useful to preserve progress in a form that is mathematically organized and epistemically explicit.

The central aim of the present work is therefore modest but important. We do not claim that every constant and every effective interaction has already been uniquely derived from first principles. Instead, we provide a framework-consolidation paper that records what is currently established, what is derived only within reduced constructions, what is empirically fixed, and what remains open. The main unresolved issue highlighted is the microscopic bridge from the base cone geometry to the reduced mechanisms used later in the paper.

2. Foundational Cone Geometry

2.1 Static cone background

The starting point of the framework is a static cone geometry of the form: $ds^2 = dr^2 + \beta^2 r^2 d\phi^2 + dz^2$ with deficit angle: $\delta = 2\pi(1 - \beta)$ It is convenient to associate the quantity $Q = 1 - \beta$ with the topological strength of the defect. At this level, the cone geometry is an ansatz defining the basic background structure of the model.

2.2 Pause scale and regime transition

A central ingredient of the framework is the introduction of a characteristic scale: $r_\square = 1 / \mu$ interpreted as a pause or crossover scale between compressive and tensile regimes. This scale is structural at the framework level, even though the full microscopic derivation of the parameter μ remains open.

2.3 Asymptotic scaling relation

One of the principal motivations for the cone-sector construction is the emergence, within the framework, of a quarter-power asymptotic relation of the form: $v_\infty \approx (\Lambda^2 G M \mu)^{1/4}$ This relation is intended to arise from a combination of tension-balance and phase-synchronization reasoning in the asymptotic regime. The precise numerical values of the parameters entering this relation are not all fixed by the base geometry alone.

Status label: Framework-level / partially derived / partly calibrated.

3. Reduced Sector EFT and Exchange-Odd Coupling

3.1 Parent mediator over the sector interval

To describe effective interactions between reduced sector variables, we introduce a parent mediator $\Phi(\psi, \mathbf{x})$ defined over the angular sector interval $[\psi_1, \psi_2]$. We consider the parent interaction: $\text{Sint}_{\text{parent}} = \int d^d \mathbf{x} X^2(\mathbf{x}) \int_{\psi_1}^{\psi_2} d\psi J(\psi) \Phi(\psi, \mathbf{x})$ where $X(\mathbf{x})$ denotes the sourcing field and $J(\psi)$ is a kernel encoding the angular weighting of the interaction.

3.2 Sector projection

In the sharp-projector limit, we define endpoint sector fields by: $Y_a(\mathbf{x}) = \Phi(\psi_1, \mathbf{x})$, $Y_b(\mathbf{x}) = \Phi(\psi_2, \mathbf{x})$ This projection isolates the two endpoint sectors of the interval and yields an effective two-field description. Smoother projection prescriptions may ultimately be preferable, but the sharp-projector limit is sufficient for the present reduced derivation.

3.3 Endpoint-antisymmetric kernel choice

To obtain an exchange-odd coupling, we choose a kernel whose endpoint values are equal and opposite. One convenient choice is: $J(\psi) = \lambda_0 (\sin \psi - (\sin \psi_1 + \sin \psi_2) / 2)$ Then the endpoint couplings are $\lambda_a = J(\psi_1)$ and $\lambda_b = J(\psi_2)$, with $\lambda_a = -\lambda_b$. The resulting effective interaction becomes: $V_{i\square\square} = X^2 (\lambda_a Y_a + \lambda_b Y_b)$ Introducing the symmetric and antisymmetric combinations $Y_+ = (Y_a + Y_b)/\sqrt{2}$ and $Y_- = (Y_a - Y_b)/\sqrt{2}$, we obtain: $V_{i\square\square} = \sqrt{2} \lambda X^2 Y_-$ for an appropriate effective

coupling λ .

3.4 Interpretation and caveat

This establishes the desired exchange-odd interaction structure within the parent-kernel / sharp-projector reduction, but it should not be read as a complete microscopic derivation from the full cone theory.

Status label: Derived within reduced model.

4. Exchange Symmetry and Remaining Sector Conditions

4.1 Exchange symmetry requirement

The reduced-model symmetry requirement: $\mathbf{S}[\mathbf{Y}_a, \mathbf{Y}_b, \mathbf{X}] = \mathbf{S}[\mathbf{Y}_b, \mathbf{Y}_a, \mathbf{X}]$ motivates the suppression of certain odd sector self-interactions, including terms such as $\mathbf{Y}^{-3}$, but does not by itself constitute a full microscopic derivation.

4.2 Current status and mode splitting

The reduced EFT supports the exchange-symmetry logic, but sector exchange does not automatically imply an intrinsic field parity $\mathbf{Y} \rightarrow -\mathbf{Y}$. Therefore, the suppression of odd self-interactions should presently be regarded as motivated within the reduced model. To project to the odd mediator, we desire a quadratic structure of $\mathbf{V}^{(2)} = \frac{1}{2} \mathbf{m}_+^2 \mathbf{Y}_+^2 + \frac{1}{2} \mathbf{m}_-^2 \mathbf{Y}_-^2$ where $\mathbf{m}_+^2 \gg \mathbf{m}_-^2$, which remains open microscopically.

Status label: Open / reduced-model motivated.

5. First-Principles Derivation of the Scaling Exponent

The empirical scaling law $\gamma = 0.144 \pm 0.060$ (from SPARC data) has previously been treated as a target for tuning dynamical parameters. In this section, we derive the scaling exponent from first principles using two independent but complementary methods: the one-loop Renormalization Group (RG) flow of the geometric mismatch field, and the geometric eigenvalue problem on the physical galactic junction.

5.1 Minimal two-field defect EFT and anomalous dimension

The defect sector contains a localized mismatch mode $\mathbf{X}$, identified with the low-lying inter-sector mass mismatch, and a mediator mode $\mathbf{Y}$ that couples cubically to $\mathbf{X}$. The minimal Euclidean EFT in $d = 6 - \epsilon$ is: $\mathcal{L}_{\text{def}} = \frac{1}{2}(\partial\mathbf{X})^2 + \frac{1}{2}(\partial\mathbf{Y})^2 + (\mathbf{g}_1/2)\mathbf{X}^2\mathbf{Y} + (\mathbf{g}_2/6)\mathbf{Y}^3$. Solving $\beta_1 = \beta_2 = 0$ reveals that the formal mixed fixed point is nonphysical, as both squared couplings become negative. The physically relevant perturbative branch is the reduced theory with $\mathbf{g}_2 = 0$. In that sector, the fixed point is: $\mathbf{g}_1^* = (4\pi)^3 \epsilon (4/13)$. At the fixed point, the $\mathbf{X}$ -field anomalous dimension is $\eta_{\mathbf{X}}^* = 2\epsilon/13 + \mathcal{O}(\epsilon^2)$. Specializing formally to $\epsilon = 1$ (the five-dimensional case) gives the central prediction: $\eta_{\mathbf{X}}^* \approx 2/13 \approx 0.154$. Consequently, the scaling dimension is $\Delta_{\mathbf{X}} = (d - 2 +$

$$\eta_{x*})/2 = 24/13 \approx 1.846.$$

5.2 Geometric Eigenvalue on the $d\Box = 3$ Cone Sector

Independently, we compute the dimension of the transmission operator $\mathbf{O}_\mu$ spanning the galactic junction ($d\Box = 3$). The field lives on an angular sector $[\psi_1, \psi_2]$ corresponding to the geometric cone, with physically distinct mixed boundary conditions: an inner baryonic boundary at $\psi_1 = 35^\circ$ (Neumann: $\Phi' = 0$) and an outer sovereign boundary at $\psi_2 = 70^\circ$ (Dirichlet: $\Phi = 0$). Numerical evaluation of the Legendre determinant yields an exact geometric eigenvalue of: $\mathbf{v}_0 \approx 2.31$ For a boundary operator on a $d\Box = 3$ defect, the scaling dimension is exactly $\Delta_\mu = \mathbf{v}_0 \approx 2.31$.

5.3 Geometric estimate of the sourcing multiplicity

A second quantity entering the phenomenological matching is the effective sourcing multiplicity $\mathbf{n}$. For the defect sector interval $\Delta\psi = 35^\circ = 7\pi/36$, a natural dimensionless winding count associated with the embedding of this interval into the full azimuthal period is: $\mathbf{N_wind} = 2\pi / \Delta\psi = 72/7 \approx 10.286$ An image-based counting argument suggests that the mixed boundaries reduce the effective winding count by two units. Truncating to the integer part and applying this reduction yields: $\mathbf{n_eff} \approx \lfloor \mathbf{N_wind} \rfloor - 2 = 10 - 2 = 8$

5.4 Synthesis and Epistemic Conclusion

The predicted scaling exponent γ relates the transmission operator dimension Δ_μ to the mismatch field dimension Δ_x via the sourcing relation $\mathbf{X} \sim \Sigma^{b^{1/n}}$: $\gamma = \Delta_\mu / (\mathbf{n} \cdot \Delta_x) = 2.31 / (8 \times 1.846) \approx 0.156$ This predicted value lies within 0.2σ of the empirical center of the SPARC data range ($\gamma = 0.144 \pm 0.060$).

6. Derivational Status Audit

Table 1. Current derivational status of the cone-sector program

Item	Result / Claim	Status	Basis / Caveat
Cone geometry	Static metric $ds^2 = dr^2 + \beta^2 r^2 d\phi^2 + dz^2$	Established framework	Base geometric ansatz.
Topological param	$Q = 1 - \beta$	Established framework	Physical interpretation fixed.
Pause scale	$r\Box = 1 / \mu$	Framework-level	Micro-derivation of μ open.
Asymptotic scaling	$v^\infty \approx (\Lambda^{*2} G M \mu)^{1/4}$	Framework-level	Constants not fully derived.
Odd coupling	$V_{i\Box\Box} \propto X^2 Y_-$	Derived (Reduced Model)	Kernel/projector dependent.
Exchange symmetry	Suppresses Y_-^3	Open microscopically	EFT logic clear, full proof open.
Effective n	$\mathbf{n} = 8$	Derived (Geometric Est)	Boundary exclusion model.

7. RG/RDG Extension Program

The broader RG/RDG program is intended to deepen the present framework by embedding the static cone construction into a more dynamical setting. Prospective targets for derivation include the Laurent $1/x^2$ term, vertical thickness, and axion spectra.

8. Conclusion

We have presented a framework-consolidation paper linking conical geometry, effective sector interactions, and winding-based multiplicity models. The achievement here is that γ is no longer a fitted parameter, but an observable constrained by verifiable field-theoretic and geometric topologies.

Appendices

Appendix A. Renormalization Details: Fixed point $g_{\text{r}}^{*2} = (4\pi)^3 \varepsilon (4/13)$ leads to $\eta_{\text{x}}^* = 2\varepsilon/13$.

Appendix B. Multiplicity Model: $N_{\text{wind}} = 10.28$. Image mirror reduction = 2. Result: $n = 8$.

Appendix C. Notation: β = cone param; Q = deficit; μ = inverse pause scale; Λ^* = asymptotic scale; n = multiplicity.